

Research Statement

Compositional Hybrid Digital Twins for Fermented Microbial Ecosystems

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1 Introduction and Motivation

Fermented food systems are complex microbial ecosystems characterized by non-linear, evolving interactions between species and their environments. Predictive models of these systems are essential for food innovation and safety, but they are often difficult to capture using a single modeling paradigm. This research statement proposes a **Compositional Hybrid Digital Twin** framework that systematically integrates mechanistic ecological models (e.g., consumer-resource models) with machine learning surrogates, enabling flexible and scalable simulation of biological ecosystems.

2 Current Research: Hybrid Physics-Data Modeling

In my current work at Sonatrach, I have developed a **Physics-Informed Digital Twin** for gas turbine monitoring. My research focuses on the hybridization of mechanistic physical laws (thermodynamics) with data-driven architectures (Mamba-SSM/xLSTM). I have implemented custom **PINN loss functions** that enforce physical consistency in generative surrogates. This experience has given me a deep understanding of how to blend "first-principles" equations with partially understood observed dynamics—a challenge that is directly relevant to modeling the evolving interactions within microbial communities.

3 Proposed Research Directions

Building upon the objectives of the **Catalyst (Reliable Hybrid Model Forge)** project, I propose to investigate the following axes within the context of fermentation ecosystems:

3.1 1. Composable Hybridization Operators

I intend to investigate formal operators of model composition that enable the systematic integration of mechanistic ecological models and machine learning components. By designing well-defined hybridization operators, we can learn corrections to partially specified biological dynamics or infer latent microbial processes from incomplete experimental observations.

3.2 2. Reusable Hybrid Modeling Patterns

The core of my research will focus on defining reusable patterns for microbial hybridization (e.g., surrogate replacement, residual learning, multi-fidelity switching). These patterns will allow for a "modular" approach to digital twin construction, where modeling strategies can be adapted as new biological knowledge or datasets become available.

3.3 3. Multi-Fidelity Model Blending

I propose to investigate adaptive switching mechanisms between models of different complexity, such as consumer-resource models, Lotka-Volterra dynamics, and neural surrogates. By leveraging my background in uncertainty quantification, I aim to ensure that the digital twin automatically selects the most appropriate modeling fidelity based on the available data and the required precision.

4 Alignment with the EDT Program

My background in hybrid modeling and industrial implementation aligns with the goal of creating a national infrastructure for digital twin engineering. I am eager to contribute to the ****Artemis platform**** by providing a compositional modeling layer that can handle the unique challenges of biological and microbial digital twins across the consortium.

5 Conclusion

The goal of my PhD will be to move biological digital twins from ad-hoc simulations to a rigorous, compositional hybrid framework. By bridging the gap between computational biology, ecological modeling, and machine learning, I aim to provide the foundational tools necessary for the next generation of reliable and scalable microbial digital twins.